

Continuous Time Dynamics

Optimization and Learning for Robot Control

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General Nonlinear System

The most general smooth nonlinear system:

$$\dot{x}(t) = f(x(t), u(t), t)$$

where

$$x(t) \in \mathbb{R}^n \quad (\text{state})$$

$$u(t) \in \mathbb{R}^m \quad (\text{input})$$

State derivative defines system evolution.

Mechanical System State Representation

For mechanical systems:

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

where

$$q \in \mathbb{R}^n \quad (\text{configuration})$$

Thus state dimension is:

$$x \in \mathbb{R}^{2n}$$

Configuration space: $q \in \mathcal{Q}$

Example: Pendulum

Equation of motion:

$$m\ell^2\ddot{\theta} + mg\ell \sin \theta = \tau$$

Define state:

$$x_1 = \theta, \quad x_2 = \dot{\theta}$$

Then:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{1}{m\ell^2} (\tau - mg\ell \sin x_1)$$

Nonlinear due to $\sin x_1$.

Control-Affine Systems

Many nonlinear systems can be written as:

$$\dot{x} = f(x) + G(x)u$$

where:

$f(x)$ (drift dynamics)

$G(x)$ (input matrix)

This is called **control-affine form**.

Pendulum in Control-Affine Form

From:

$$\dot{x}_2 = \frac{1}{m\ell^2} (\tau - mg\ell \sin x_1)$$

Rewrite:

$$\dot{x} = \begin{bmatrix} x_2 \\ -\frac{g}{\ell} \sin x_1 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m\ell^2} \end{bmatrix} \tau$$

Thus:

$$f(x) = \begin{bmatrix} x_2 \\ -\frac{g}{\ell} \sin x_1 \end{bmatrix}, \quad G(x) = \begin{bmatrix} 0 \\ \frac{1}{m\ell^2} \end{bmatrix}$$

Manipulator Dynamics

General rigid-body manipulator:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = B\tau$$

where

$M(q)$ (inertia matrix)

$C(q, \dot{q})\dot{q}$ (Coriolis/centrifugal)

$g(q)$ (gravity)

State-Space Form of Manipulator

Define:

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

Then:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = M^{-1}(q)(B\tau - C(q, \dot{q})\dot{q} - g(q))$$

Thus:

$$\dot{x} = f(x) + G(x)\tau$$

All mechanical systems admit control-affine form.

Linear time-invariant (LTI) system:

$$\dot{x} = Ax + Bu$$

Time-invariant if:

$$A(t) = A, \quad B(t) = B$$

Otherwise time-varying.

Linearization of Nonlinear Systems

Given nonlinear system:

$$\dot{x} = f(x, u)$$

Linearize around equilibrium (x_0, u_0) :

$$\delta\dot{x} = \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0} \delta x + \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0} \delta u$$

Define:

$$A = \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0}$$

Then:

$$\delta\dot{x} = A\delta x + B\delta u$$

Key Message

- All mechanical systems:

$$\dot{x} = f(x) + G(x)u$$

- Linear systems are special case.
- Nonlinear systems are locally approximated by linear ones.